

Project Design Phase

Solution Architecture

Date	01 October 2026
Team ID	SWTID-2026-7951
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	4 Marks

The supplied architecture template defines solution architecture as the bridge between business problems and technology solutions, including structure, behaviour, features, development phases and solution requirements.

Solution Architecture & Data Flow

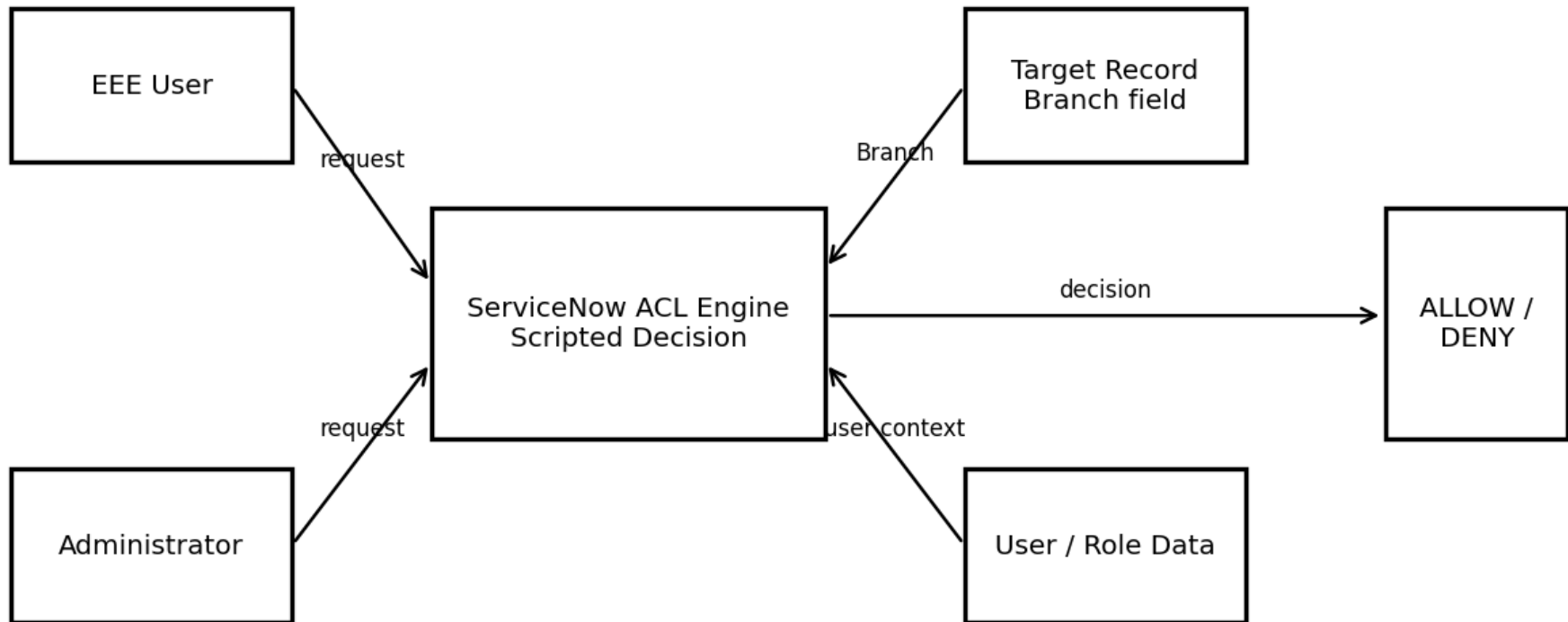


Figure 1:

Architecture and data flow of the Script-Controlled ACL solution.

Component	Role	Input	Output
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EEE User	Requests a record.	User identity + record request.	Record if branch condition passes.
Administrator	Performs unrestricted management access.	Admin role + record request.	Record access across branches.
ACL Engine	Runs scripted authorization.	User role/branch + record Branch.	Allow or deny.
Target Record	Provides Branch value.	Requested record.	Branch used for decision.
User / Role Data	Provides authorization context.	Logged-in user identity and role/branch.	Context for ACL.

Access Decision Flow:

1. User requests a record.
2. ACL identifies the logged-in user and role/branch context.
3. ACL reads the record's Branch value.
4. Administrator → allow.
5. Otherwise, EEE user + Branch = EEE → allow.
6. Otherwise → deny access.

Design characteristics:

record-level authorization, explicit administrator exception, simple conditional logic, maintainability, and extensibility to additional branches.